RKNN3 SDK Release Note

Document ID: RK-JC-YF-429

Version: V1.0.4

Date: 2026-05-15

Confidentiality: □Top Secret □Secret □Internal ■Public

Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian,PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Target Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers

Revision History

Version	Author	Date	Description	Approved By
V0.2.0	HPC	2025-08-16	Initial version	Vincent
V0.3.0b0	HPC	2025-09-12	1. Updated the list of supported models 2. Updated accuracy and performance data	Vincent
v0.4.0b0	HPC	2025-11-03	1. Update the list of supported models 2. Update the overview; model accuracy, and performance data	Vincent
V1.0.0	HPC	2026-01-23	1. Updated the list of supported models 2. Updated accuracy and performance data	Vincent
V1.0.4	HPC	2026-05-15	1. Updated the list of supported models 2. Updated accuracy and performance data 3. Updated supported hardware platforms	Vincent

Table of Contents

---

1 Overview

The RKNN3 SDK provides the software stack required to deploy AI models to the RK1820/RK1828 and the RK3572 platform, including a PC development kit (RKNN3 Toolkit), on-device runtime API (RKNN3 Runtime), and model conversion and deployment examples (RKNN3 Model Zoo). This SDK release supports both coprocessor mode (RK1820/1828) and non-coprocessor mode (RK3572). The main differences between the two modes are as follows:

> **[Coprocessor Mode (RK182X)]**

- **Host SoC**: Acts as the system's core, responsible for task scheduling, resource allocation, and overall control.
- **RK1820/RK1828 Coprocessor**: Serves as an AI computation acceleration unit, focusing on high-performance neural network inference tasks.
- **PCIe/USB High-Speed Interface**: Enables low-latency, high-bandwidth data interaction between the host and the coprocessor.

> **[Non-Coprocessor Mode (RK3572)]**

- **RK3572 SoC**: An SoC platform with an integrated high-performance NPU, supporting direct execution of AI inference tasks.
- **Unified Memory Architecture**: The NPU shares memory space with the CPU/GPU.

1.1 Supported Hardware Platforms

- RK1820/RK1828 Coprocessor
- RK3572

1.2 Supported Systems

- Android/Linux/Windows

2 Key Features and Enhancements

- Added RK3572 platform support (Beta)
- Added Windows platform support
- Added Ethernet communication type support
- Added device sleep/wake-up support
- Added server-side automatic communication mode selection
- Added streaming weight loading support
- Added model encryption support
- Added LoRA support
- Added session pause/resume functionality
- Added session input callback functionality
- Added multi-session parallel execution support
- Added KVCache import/export interface
- Added component version compatibility validation

3 Supported Models

The currently supported models are listed below:

Model Name	Model Source
Qwen2.5-0.5B	https://huggingface.co/Qwen/Qwen2.5-0.5B
Qwen2.5-3B	https://huggingface.co/Qwen/Qwen2.5-3B-Instruct
Qwen2.5-7B	https://huggingface.co/Qwen/Qwen2.5-7B-Instruct
Qwen3-0.6B	https://huggingface.co/Qwen/Qwen3-0.6B
Qwen3-1.7B	https://huggingface.co/Qwen/Qwen3-1.7B
Qwen3-4B	https://huggingface.co/Qwen/Qwen3-4B
Qwen3-8B	https://huggingface.co/Qwen/Qwen3-8B
HY-MT1.5-1.8B	https://huggingface.co/tencent/HY-MT1.5-1.8B
Youtu-LLM-2B	https://huggingface.co/tencent/Youtu-LLM-2B
GLM-Edge-1.5B-Chat	https://modelscope.cn/models/ZhipuAI/glm-edge-1.5b-chat
Qwen2.5-VL-3B	https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct
Qwen2.5-VL-7B	https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct
Qwen2.5-Omni-3B (Thinker)	https://huggingface.co/Qwen/Qwen2.5-Omni-3B
Qwen3-VL-2B	https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct
Qwen3-VL-4B	https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct
FastVLM	https://github.com/apple/ml-fastvlm
InternVL3-2B	https://huggingface.co/OpenGVLab/InternVL3-2B
InternVL3_5-4B	https://huggingface.co/OpenGVLab/InternVL3_5-4B-Instruct
MiMo-VL-7B-RL	https://huggingface.co/XiaomiMiMo/MiMo-VL-7B-RL
Gemma4-2B	https://huggingface.co/google/gemma-4-E2B-it
Gemma4-4B	https://huggingface.co/google/gemma-4-E4B-it
SmolVLM-500M-Instruct	https://huggingface.co/HuggingFaceTB/SmolVLM-500M-Instruct
SmolVLM2-500M-Video-Instruct	https://huggingface.co/HuggingFaceTB/SmolVLM2-500M-Video-Instruct
UI-TARS-2B-SFT	https://huggingface.co/ByteDance-Seed/UI-TARS-2B-SFT
PaddleOCR VL	https://huggingface.co/PaddlePaddle/PaddleOCR-VL

Model Name	Model Source
Qwen3-Reranker-0.6B	https://huggingface.co/Qwen/Qwen3-Reranker-0.6B
Qwen3-Reranker-4B	https://huggingface.co/Qwen/Qwen3-Reranker-4B
Qwen3-Embedding-0.6B	https://huggingface.co/Qwen/Qwen3-Embedding-0.6B
Qwen3-Embedding-4B	https://huggingface.co/Qwen/Qwen3-Embedding-4B
gme-Qwen2-VL-2B-Instruct	https://huggingface.co/Alibaba-NLP/gme-Qwen2-VL-2B-Instruct
Qwen3-ASR-0.6B	https://huggingface.co/Qwen/Qwen3-ASR-0.6B
Qwen3-TTS-12Hz-1.7B	https://huggingface.co/Qwen/Qwen3-TTS-12Hz-1.7B-Base
VITS	https://github.com/jaywalnut310/vits
Whisper	https://huggingface.co/openai/whisper-large-v3
SenseVoiceSmall	https://modelscope.cn/models/iic/SenseVoiceSmall
Zipformer	https://huggingface.co/pfluo/k2fsa-zipformer-chinese-english-mixed
SigLIP	https://huggingface.co/google/siglip-so400m-patch14-384
Siglip2-so400m	https://huggingface.co/google/siglip2-so400m-patch14-384
MetaCLIP2	https://huggingface.co/facebook/metaclip-2-worldwide-m16-384
Dinov3	https://huggingface.co/facebook/dinov3-vits16-pretrain-lvd1689m
Depth-Anything-V2-small	https://huggingface.co/depth-anything/Depth-Anything-V2-Small
GR00T-N1.6-3B	https://huggingface.co/nvidia/GR00T-N1.6-3B
MobilenetV1	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/mobilenet_v1/mobilenet_v1_1.0_224.tflite
MobilenetV2	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/mobilenet/mobilenetv2-12.onnx
Resnet50V2	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/resnet/resnet50-v2-7.onnx
YOLOv5s	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/yolov5/yolov5s_rknn3.onnx
YOLOv6s	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/yolov6/yolov6s_rknn3.onnx
YOLOv8s	https://ftrg.zbox.filez.com/v2/delivery/data/95f00b0fc900458ba134f8b180b3f7a1/examples/yolov8/yolov8s_rknn3.onnx

Users can download pre-converted RKNN models from the following cloud drive: RKNN3_SDK (https://console.box.lenovo.com/l/H1fig1, Access Code: rknn). The specific path is as follows: `RKNN3_SDK/rknn3_models/v1.0.4`

4 Model Performance

This section shows the performance of typical LLM, VLM, full-modal and CNN models on the RK1820/RK1828 coprocessor.

- **LLM Model Performance**

Model Name	Accelerator	Input Tokens	New Tokens	TTFT (ms)	TPOT (ms)	Decode TPS
Qwen2.5-0.5B	RK182X	128	128	22.74	4.48	223.40
Qwen2.5-1.5B	RK182X	128	128	49.14	6.69	149.39
Qwen2.5-3B	RK182X	128	128	85.54	9.69	103.24
Qwen2.5-7B	RK1828	128	128	162.25	14.19	70.47
Qwen3-0.6B	RK182X	128	128	28.61	5.49	182.26
Qwen3-1.7B	RK1828	128	128	54.34	7.17	139.39
Qwen3-4B	RK1828	128	128	109.78	11.30	88.47
Qwen3-8B	RK1828	128	128	182.20	16.30	61.34

- **VLM Model Performance**

Model	Accelerator	Vision Resolution	Vision(ms)	LLM TTFT (ms)	LLM Decode TPS
FastVLM_1.5B_stage3	RK182X	512 * 512	168.85	49.83	151.01
InternVL3-2B	RK182X	448 * 448	184.19	49.85	147.62
InternVL3_5-4B	RK1828	448 * 448	176.99	110.06	87.95
Qwen2.5-VL-3B	RK182X	392 * 392	231.4	97.85	51.48
Qwen2.5-VL-3B	RK1828	392 * 392	212.28	87.6	104.05
Qwen2.5-VL-7B	RK1828	392 * 392	215.63	163.5	70.00
Qwen3-VL-2B	RK182X	384 * 384	114.38	56.55	142.00
Qwen3-VL-4B	RK1828	384 * 384	117.6	111.45	87.8
MiMo-VL-7B-RL	RK1828	392 * 392	216.56	173.59	64.97
MiniCPM_V_4	RK1828	448 * 448	236.67	97.81	106.56

- **Full-Modal Model Performance**

Model Name	Accelerator	Vision Resolution	Vision(ms)	Audio(ms)	LLM TTFT (ms)	LLM Decode TPS
Qwen2.5-Omni-3B (Thinker)	RK1828	392 * 392	220.01	93.60	86.96	104.01
Gemma-4-E2B	RK182X	384 * 384	62.20	103.98	99.41	70.19
Gemma-4-E4B	RK1828	384 * 384	77.72	119.82	169.93	51.02

- **CNN Model Performance**

Model Name	Accelerator	Resolution	Single-Core Performance (fps)	Multi-Batch Multi-Core Performance (fps)
MobilenetV1	RK182X	224 * 224	388.41	1501.34
MobilenetV2	RK182X	224 * 224	279.67	1290.93
Resnet50V2	RK182X	224 * 224	112.58	843.24
YOLOv5s	RK182X	640 * 640	34.54	214.49
YOLOv6s	RK182X	640 * 640	30.73	203.46
YOLOv8s	RK182X	640 * 640	33.01	212.32

Note:

1. "RK182X" in the tables indicates that the accelerator chip can be either RK1820 or RK1828.
2. For `Qwen2.5-VL-3B`:

- When using RK1820 as the accelerator, a two-stage scheme is adopted (the LMHead runs on RK3588).
- When using RK1828 as the accelerator, the model runs entirely on the coprocessor.

3. The RK1820/RK1828 coprocessor NPU frequency is 1GHz.
4. Tests are based on an RK3588 + RK1820/RK1828 setup connected via PCIe, with the RK3588 set to performance mode.
5. TTFT: Time To First Token.
6. TPOT: Time Per Output Token.
7. TPS: Tokens Per Second.
8. The Vision and LLM timeouts of VLM were tested independently, and the Input Tokens and New Tokens of the LLM part were both set to 128.

5 Model Accuracy

- **LLM Model Accuracy**

Model Name	Accelerator	Dataset	Orig Model Acc (float32)	RKNN3 Acc (W4A16 G32)
Qwen2.5-0.5B	RK182X	gsm8k	40.71	36.09
Qwen2.5-3B	RK182X	gsm8k	79.91	80.67
Qwen3-4B	RK1828	gsm8k	90.6	89.84

- **VLM Model Accuracy**

Model Name	Dataset	Orig Model Acc (float32)	RKNN3 Acc (W4A16 G32)
FastVLM_1.6B	MMbench(cn)	58.42	60.48
Qwen2.5-VL-3B	MMbench(cn)	76.8	75.43
Qwen2.5-VL-7B	MMbench(cn)	79.98	81.19
InternVL3_2B	MMbench(cn)	77.23	72.51
InternVL3_5-4B	MMbench(cn)	78.69	77.75
mimo_vl_7b	MMbench(cn)	74.7	69.85

- **CNN Model Accuracy**

Model Name	Dataset	Orig Model Float32 (TOP1)	Orig Model Float32 (TOP5)	RKNN3 W8A8 (TOP1)	RKNN3 W8A8 (TOP5)
MobilenetV1	imagenet	0.677	0.877	0.676	0.876
MobilenetV2	imagenet	0.694	0.888	0.680	0.882
Resnet50V2	imagenet	0.729	0.911	0.721	0.906

Model Name	Dataset	Orig Model Float32 AP@0.5:0.95	Orig Model Float32 AP@0.5	RKNN3 W8A8 AP@0.5:0.95	RKNN3 W8A8 AP@0.5
YOLOv5s	coco2017	0.326	0.481	0.310	0.471
YOLOv6s	coco2017	0.403	0.551	0.385	0.534
YOLOv8s	coco2017	0.39	0.525	0.380	0.513

Note:

1. W4A16 G32 means that weights use 4-bit asymmetric quantization and activations use 16-bit floating point representation. Quantization parameters are assigned for every group of 32 weights along the input channel dimension.
2. W8A8 means that both weights and activations use 8-bit asymmetric quantization.

6 Recommended Server Configuration

The recommended server configurations and conversion time estimates are as follows:

Model Name	Recommended Server Configurations	Estimated Conversion Time
Qwen 2.5 0.5B	32-core CPU / 8 GB RAM / 1 TB SSD/HDD	~11 minutes
Qwen 2.5 1.5B	32-core CPU / 16 GB RAM / 1 TB SSD/HDD	~26 minutes
Qwen 2.5 3B	32-core CPU / 32 GB RAM / 1 TB SSD/HDD	~52 minutes
Qwen 2.5 7B	32-core CPU / 64 GB RAM / 1 TB SSD/HDD	~105 minutes

- If you encounter insufficient memory, you can try enabling a Swap partition to expand memory. See reference: https://wiki.debian.org/Swap.

7 References

7.1 Performance Testing Methods

- **CNN Model Performance Testing**

  Reference: **rknn3-runtime/examples/rknn3_model_test_demo/README_CN.md**

- **LLM Model Performance Testing**

  Reference: **rknn3-model-zoo/tools/rknn3_llm_test/README.md**

7.2 Accuracy Testing Methods

- **CNN Model Accuracy Testing**

  The rknn3-model-zoo integrates testing methods and code for CNN model accuracy. Users who need to re-verify model accuracy can refer to the instructions shown below.

  1. Classification models: **rknn3-model-zoo/examples/mobilenet_v2/README.md**
  2. Detection models: **rknn3-model-zoo/examples/yolov8/README.md**

- **LLM Model Accuracy Testing**

  The rknn3-model-zoo also integrates methods and code for LLM model accuracy testing, with support for the CMMLU dataset. For specific testing procedures, refer to **rknn3-model-zoo/tools/rknn3_llm_test/README.md**